

RST2016A

1.0 Specification References

Parameter	Description
a. Rakon part number	T6302
b. Description	40.0 MHz RST2016A VCTCXO
c. Package	L x W: 2.0 x 1.6 mm. H = 0.8 mm max.



2.0 Absolute Maximum Rating ¹

Parameter	Min.	Max.	Unit
a. Power supply	-0.6	+4.6	V
b. Storage temperature	-40	85	°C

3.0 Frequency Characteristics

Parameter	Min.	Typ.	Max.	Unit	Test Condition / Description
a. Nominal frequency		40.0		MHz	
b. Frequency calibration			±1	ppm	Offset from nominal frequency measured at 25°C ±2°C
c. Reflow shift			±1	ppm	Two consecutive reflows as per attached profile after 2 hours relaxation at 25°C
d. Temperature range	-40		85	°C	The operating temperature range over which the frequency stability is measured
e. Frequency stability over temperature			±0.5	ppm	Referenced to the midpoint between minimum and maximum frequency value over the specified temperature range ² Control voltage set to midpoint of control voltage
f. Frequency slope			±0.1	ppm/°C	Minimum of one frequency reading every 2°C over the operating temperature range ²
g. Static temperature hysteresis			0.6	ppm	Frequency change after reciprocal temperature ramped over the operating range. Frequency measured before and after at 25°C
h. Sensitivity to supply voltage variations			±0.1	ppm	Supply voltage varied ±5% at 25°C
i. Sensitivity to load variations			±0.2	ppm	±10% load change at 25°C ³
j. Long term stability			±1 ±5	ppm	Frequency drift over 1 year at 25°C Frequency drift over 10 years at 25°C
k. Acceleration sensitivity			1.5	ppb/g	From 10 to 1500Hz random.

4.0 Power Supply

Parameter	Min.	Typ.	Max.	Unit	Test Condition / Description
a. Supply voltage (V _{DD})		3.3		V	With a tolerance of ±5%
b. Supply current			1.7	mA	At maximum V _{DD} ³

¹ Operating beyond this limit may result in change or permanent damage to the device.

² Parts should be shielded from drafts causing unexpected thermal gradients. Temperature changes due to ambient air currents on the oscillator can lead to short term frequency drift.

³ Specified for load stated in oscillator output section at 25°C.

5.0 Control Voltage (VCO)

Parameter	Min.	Typ.	Max.	Unit	Test Condition / Description
a. Control voltage range	0.15	1.65	3.15	V	The nominal VCO value is midway between the minimum and maximum. Voltage control should not exceed the supply voltage +0.2V or GND.
b. Frequency tuning	±8		±13	ppm	Frequency shift from minimum to maximum VCO
c. Slope		7		ppm/V	Positive
d. Linearity			10	%	Deviation from straight line curve fit
e. Control voltage input resistance		500		kΩ	Measured between VCO and GND pin

6.0 Oscillator Output

Parameter	Min.	Typ.	Max.	Unit	Test Condition / Description
a. Output waveform					DC coupled clipped sine-wave ⁴
b. Output voltage level	0.8			V	At minimum supply voltage ³
c. Output load					(10kΩ // 10pF) ±10%
d. Start up time (amplitude)			0.5	ss	Within 90% of the minimum specified output level
e. Start up time (frequency)			2	ms	Within ±0.5ppm of steady state frequency

7.0 SSB Phase Noise (40.0 MHz, at 25°C)

Parameter	Typ.	Max.	Unit.	Test Condition / Description
a. 10Hz offset		-86	dBc/Hz	Values will not be 100% screened in production.
b. 100Hz offset		-114	dBc/Hz	Values will not be 100% screened in production.
c. 1kHz offset		-135	dBc/Hz	Values will not be 100% screened in production.
d. 10kHz offset		-151	dBc/Hz	Values will not be 100% screened in production.
e. 100kHz offset		-154	dBc/Hz	Values will not be 100% screened in production.

8.0 SSB Phase Noise under vibration (40.0 MHz, at 25°C) – Custom profile

Parameter	Typ.	Max.	Unit.	Test Condition / Description
f. 10Hz offset		-85	dBc/Hz	Values will not be 100% screened in production.
g. 100Hz offset		-103	dBc/Hz	Values will not be 100% screened in production.
h. 1kHz offset		-129	dBc/Hz	Values will not be 100% screened in production.
i. 10kHz offset		-148	dBc/Hz	Values will not be 100% screened in production.
j. 100kHz offset		-151	dBc/Hz	Values will not be 100% screened in production.

⁴ External AC-Coupling capacitor required. 1nF or greater recommended.

9.0 Marking

Parameter	Test Condition / Description																																																																																										
a. Type	Engraved																																																																																										
b. Line 1	[R ##M# YM] R = Rakon, ##M# = Frequency (M=MHz, e.g. 19M2=19.2MHz) ⁵ , YM = Date code*																																																																																										
c. Line 2	[o] = Pin 1, [A__XXX] = Internal code																																																																																										
d. *Date Code	<table><tr><th colspan="4">Y - Year Code</th><th colspan="2">M - Month Code</th></tr><tr><th>Code</th><th>Year</th><th>Code</th><th>Year</th><th>Code</th><th>Month</th></tr><tr><td>A</td><td>2010</td><td>N</td><td>2023</td><td>1</td><td>January</td></tr><tr><td>B</td><td>2011</td><td>O</td><td>2024</td><td>2</td><td>February</td></tr><tr><td>C</td><td>2012</td><td>P</td><td>2025</td><td>3</td><td>March</td></tr><tr><td>D</td><td>2013</td><td>Q</td><td>2026</td><td>4</td><td>April</td></tr><tr><td>E</td><td>2014</td><td>R</td><td>2027</td><td>5</td><td>May</td></tr><tr><td>F</td><td>2015</td><td>S</td><td>2028</td><td>6</td><td>June</td></tr><tr><td>G</td><td>2016</td><td>T</td><td>2029</td><td>7</td><td>July</td></tr><tr><td>H</td><td>2017</td><td>U</td><td>2030</td><td>8</td><td>August</td></tr><tr><td>I</td><td>2018</td><td>V</td><td>2031</td><td>9</td><td>September</td></tr><tr><td>J</td><td>2019</td><td>W</td><td>2032</td><td>A</td><td>October</td></tr><tr><td>K</td><td>2020</td><td>X</td><td>2033</td><td>B</td><td>November</td></tr><tr><td>L</td><td>2021</td><td>Y</td><td>2034</td><td>C</td><td>December</td></tr><tr><td>M</td><td>2022</td><td>Z</td><td>2035</td><td></td><td></td></tr></table>	Y - Year Code				M - Month Code		Code	Year	Code	Year	Code	Month	A	2010	N	2023	1	January	B	2011	O	2024	2	February	C	2012	P	2025	3	March	D	2013	Q	2026	4	April	E	2014	R	2027	5	May	F	2015	S	2028	6	June	G	2016	T	2029	7	July	H	2017	U	2030	8	August	I	2018	V	2031	9	September	J	2019	W	2032	A	October	K	2020	X	2033	B	November	L	2021	Y	2034	C	December	M	2022	Z	2035		
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10.0 Manufacturing Information

Parameter	Test Condition / Description
a. Reflow	Solder reflow processes as per profile attached
b. Packaging description	Tape and reel. Standard packing quantity is 4000 units per reel

11.0 Environmental Specification

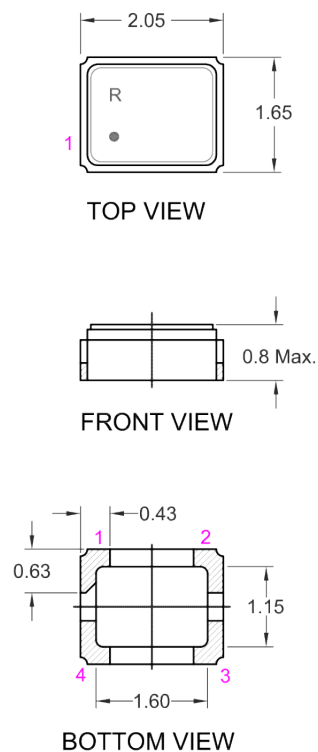
Parameter	Test Condition / Description										
a. RoHS compliant	Yes										
b. Shock	Free dropping from 150 cm height 5 times on a hard wooden board										
c. Moisture resistance	500 ±12 hours at 60°C ±3°C, 85% relative humidity ⁶										
d. Thermal cycling	The unit shall be subjected to 100 successive change of temperature cycles, then 25 ±2°C over 2 hours before testing, each cycle as below: <table> <thead> <tr> <th>Temperature</th><th>Duration:</th></tr> </thead> <tbody> <tr> <td>1. -40 +0/-6°C</td><td>30 ±3 minutes</td></tr> <tr> <td>2. 25°C ±2°C</td><td>2 – 3 minutes</td></tr> <tr> <td>3. 85 +4/-0°C</td><td>30 ±3 minutes</td></tr> <tr> <td>4. 25°C ±2°C</td><td>2 – 3 minutes</td></tr> </tbody> </table>	Temperature	Duration:	1. -40 +0/-6°C	30 ±3 minutes	2. 25°C ±2°C	2 – 3 minutes	3. 85 +4/-0°C	30 ±3 minutes	4. 25°C ±2°C	2 – 3 minutes
Temperature	Duration:										
1. -40 +0/-6°C	30 ±3 minutes										
2. 25°C ±2°C	2 – 3 minutes										
3. 85 +4/-0°C	30 ±3 minutes										
4. 25°C ±2°C	2 – 3 minutes										
e. Vibration	Frequency: 10 – 200 Hz Amplitude (total excursion): 1.5 mm (10 – 36 Hz), 4G (36 – 200 Hz) Sweep time: 1 Min./1oct 3 direction time: 2 hours for each X, Y, Z axis										

⁵ Frequency marking is only represented by the first three significant digits. For example, on an RIT2016F TCXO at 16.368MHz, its frequency code marking will be 16M3.

⁶ Frequency shift ≤2ppm after environmental conditions.

12.0 Model Outline

MODEL DRAWING

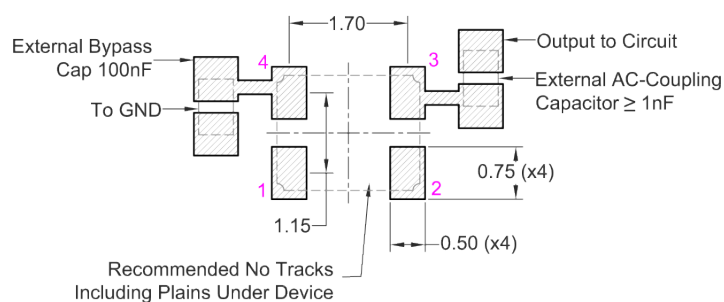


PIN CONNECTIONS

PIN	TCXO	VC-TCXO	POWER DOWN
1	GND / NC*	VCO	Enable / Disable *
2	GND	GND	GND
3	OUTPUT	OUTPUT	OUTPUT
4	VDD	VDD	VDD

* Depending on Specifications

RECOMMENDED PAD LAYOUT - Top View



TITLE: RIT/RST2016 F SERIES MODEL (Package A)

FILENAME: CAT1076

TOLERANCES:

RELATED DRAWINGS:

REVISION: B

XX =

DATE: 12-Sep-2018

X.X = ± 0.15

SCALE: 10 : 1

X.XX = ± 0.20

Millimetres

X.XXX =

X° =

Hole =

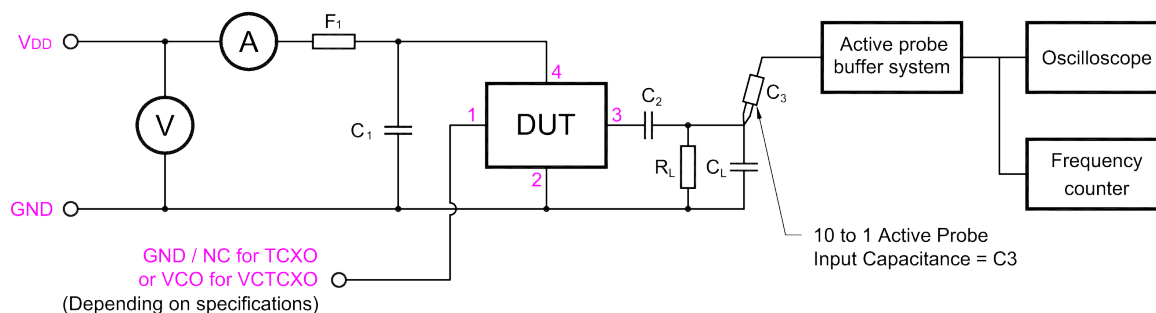


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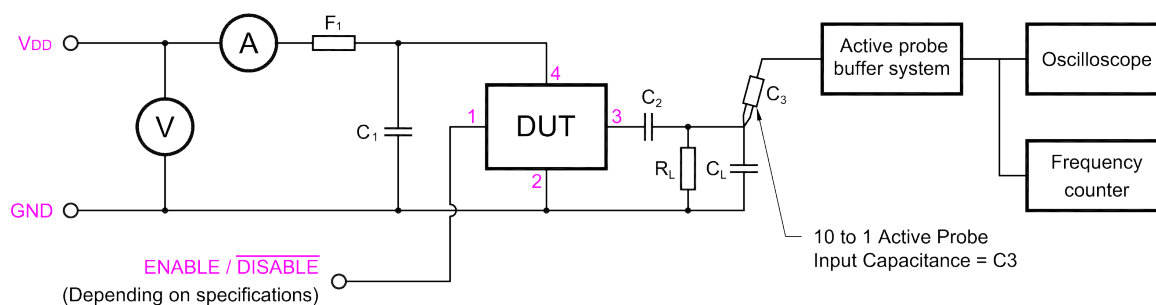
13.0 Test Circuit

CLIPPED SINEWAVE TEST CIRCUIT :

– TCXO / VCTCXO MODE



– POWER DOWN MODE



C_1 : 100nF	$C_T = C_L + C_3$ (C_3 - Oscilloscope probe capacitance)
C_2 : ≥ 1 nF	C_T as stated in OSCILLATOR OUTPUT section
R_L : 10K	F_1 : A ferrite bead or a resistor between $22\Omega \sim 47\Omega$ recommended.

TITLE: RIT/RST SERIES TCXO/VCTCXO TEST CIRCUIT (4 Pin, Package A)

FILENAME: CAT1077

RELATED DRAWINGS:

REVISION: C

DATE: 12-Sep-2018

SCALE: NTS

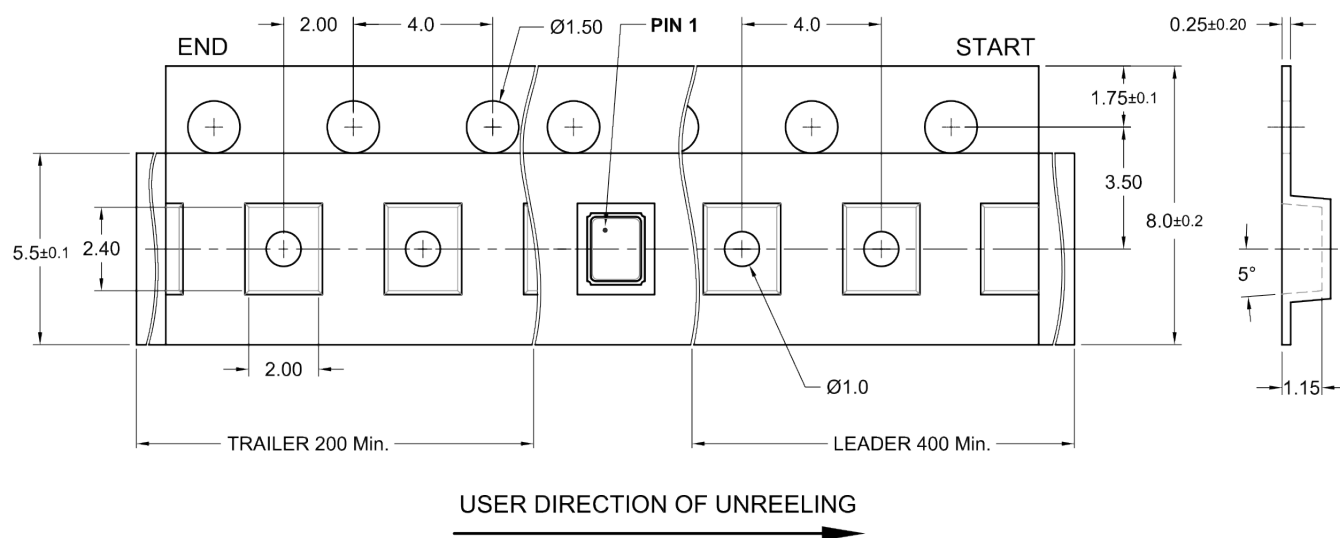
Millimetres



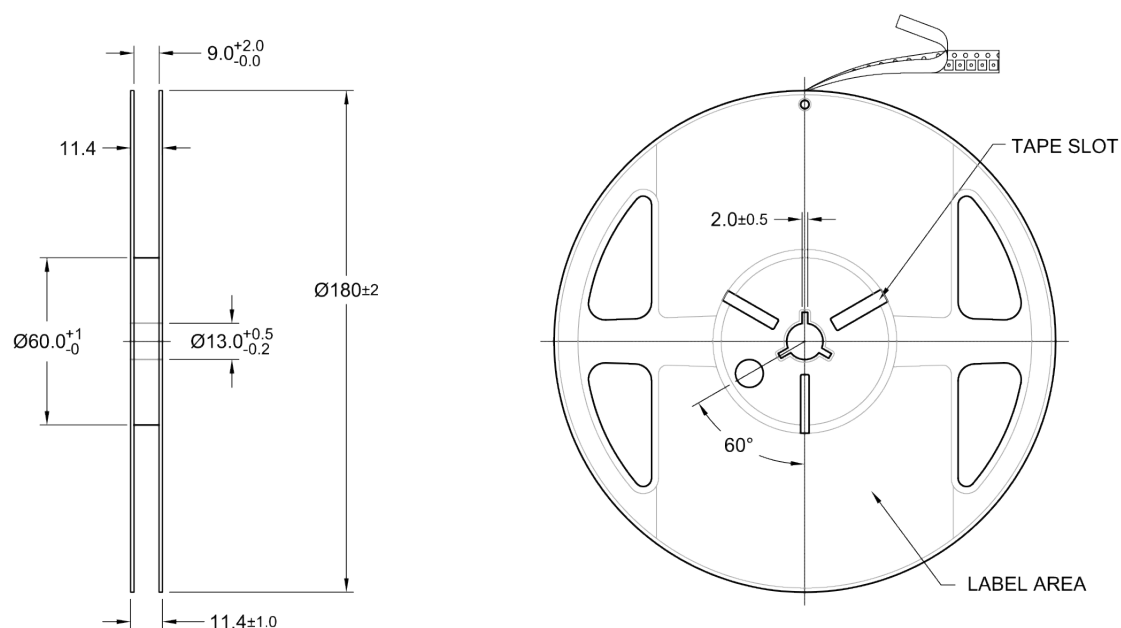
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14.0 Tape and Reel

TAPE DETAILS



REEL DETAILS



Note: The tape & reel packaging specifications follow the guidelines of the EIA Standard EIA-481.

TITLE: RST/RIT/IT 2016 TAPE & REEL (Package A, N)

RELATED DRAWINGS:

FILENAME: CAT1087

REVISION: C

DATE: 09-Jun-2020

SCALE: As above

Millimetres

TOLERANCES:

XX =

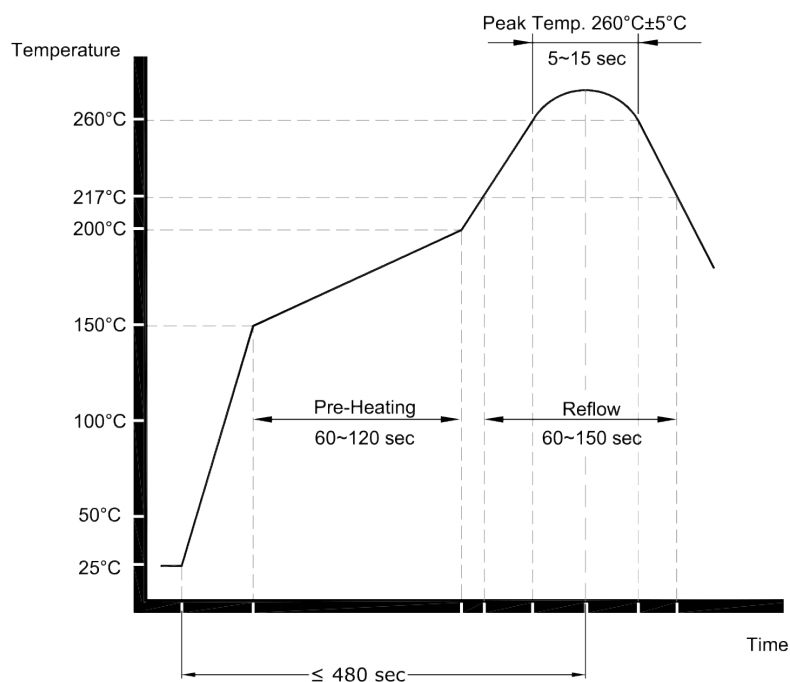
$$X.X = \pm 0.2$$

X.XX = ± 0.10

$$\begin{array}{l} X.XX \\ X^{\circ} \end{array}$$
$$\Lambda_{\text{Hole}} =$$
rakon

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15.0 Reflow



TITLE: Pb-FREE Reflow (Package A/AG)

FILENAME: CAT1036

TOLERANCES:

RELATED DRAWINGS:

REVISION: B

XX =

DATE: 03-Mar-2017

X.X =

SCALE: NTS

X.XX =

Millimetres

X.XXX =

X° =

Hole =



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16.0 Specification History

Revision	User	Notes	Approver(s)	Date
A	PK	Specification created.	JMu/ TL	2020-10-27
B	PK	Updated the phase noise to the actual values and updated the acceleration sensitivity to 1.5ppb/g	JMu/ TL	2020-11-19
C	PK	Updated laser marking and added a easy to read decode table.	JMu/ TL	2021-01-07
D	PK	Updated phase noise data to reflect degradation when VCO is present.	JMu/ TL	2021-01-15
E	PK	Updated values to the measured values.	JMu/ TL	2021-01-15
F	PK	Updated: - Frequency APR to a range between ± 8 to ± 13ppm - Supply current to 1.7mA max. - Typical Slope for the VCO to 7ppm/V (positive slope).	JMu/ TL	2021-04-08
G	PK	Changed the control voltage range to 1.65V $\pm$ 1.0V	JMu/ TL	2021-04-08
H	PK	Updated the phase noise values to include a maximum value, these values will not be 100% screened.	JMu/ TL	2021-04-12
I	PK	Updated the max. phase noise values to match the customer's requirement. Added the phase noise under vibration values. These values will not be 100% screened. Removed the typical phase noise values and only left the max. specification.	KH/ JMu/ TL	2021-04-22
J	RS	Updated product description from RIT2016F to RST2016A. Updated environmental specifications to represent testing and validation procedures.	JMu/ TL	2021-08-02
K	RS	Updated product description from TCXO to VCTCXO.	KH/ JMu/ TL	2021-08-06